

Started on	Saturday, 30 May 2026, 12:35 AM
State	Finished
Completed on	Saturday, 30 May 2026, 12:47 AM
Time taken	11 mins 38 secs
Marks	3.00/3.00
Grade	10.00 out of 10.00 (100%)

Question **1**

Correct

Mark 1.00 out of 1.00

Write a python program to find the 1-Norm of a matrix and display the results in two decimal places.

For example:

Input	Result
<code>[[-1, 3],[3, -4],[1, 7]]</code>	14.00

Answer: (penalty regime: 0 %)

```

1  '''
2  python program to find the 1-Norm of a matrix and display the results in two decimal places.
3  developed by:INIYA S
4  RegisterNumber:212225230104
5  '''
6  import os
7  os.environ["OPENBLAS_NUM_THREADS"]="1"
8  import numpy as np
9  matrix=eval(input())
10 one_matrix=np.linalg.norm(matrix,1)
11 print("{:.2f}".format(one_matrix))

```

	Input	Expected	Got	
✓	<code>[[-1, 3],[3, -4],[1, 7]]</code>	14.00	14.00	✓
✓	<code>[[1, 2, 3],[-3,-4,-1],[9,6,1]]</code>	13.00	13.00	✓

Passed all tests! ✓

► Show/hide question author's solution (Python3)

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Write a program to find L2-norm of a matrix and display the result in two decimal places.

For example:

Input	Result
[[1,2],[3,4]]	5.46
[[-1, 3],[3, -4],[1, 7]]	8.66

Answer: (penalty regime: 0. %)

Reset answer

```

1  '''
2  Program to find 2-norm of a matrix.
3  Developed by: INIYA S
4  RegisterNumber: 21225230104
5  '''
6  import os
7  os.environ["OPENBLAS_NUM_THREADS"]="1"
8  import numpy as np
9  matrix=eval(input())
10 two_matrix=np.linalg.norm(matrix,2)
11 print(round(two_matrix,2))
12
13

```

	Input	Expected	Got	
✓	[[1,2],[3,4]]	5.46	5.46	✓
✓	[[-1, 3],[3, -4],[1, 7]]	8.66	8.66	✓
✓	[[2, 3],[3, 4],[1, 8]]	9.86	9.86	✓

Passed all tests! ✓

► Show/hide question author's solution (Python3)

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Write a program to find the Infinity of a matrix and display the result in two decimal places.

For example:

Input	Result
<code>[[-1, 3],[3, -4],[1, 7]]</code>	8.00

Answer: (penalty regime: 0. %)

```

1  '''
2  program to find the Infinity of a matrix and display the result in two decimal places.
3  Developed by: INIYA S
4  RegisterNumber: 212225230104
5  '''
6  import os
7  os.environ["OPENBLAS_NUM_THREADS"]="1"
8  import numpy as np
9  matrix=eval(input())
10 inf_matrix=np.linalg.norm(matrix,np.inf)
11 print(round(inf_matrix,2))

```

	Input	Expected	Got	
✓	<code>[[-1, 3],[3, -4],[1, 7]]</code>	8.00	8.0	✓
✓	<code>[[1,2,3],[-9,-8,-3],[10,3,2]]</code>	20.00	20.0	✓

Passed all tests! ✓

► Show/hide question author's solution (Python3)

Correct

Marks for this submission: 1.00/1.00.